Math 520, F15 **Name:** Answer

Quiz 12 **SSN:** xxx-xx-

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer.

Problem 1. (10 points) Find the general solution of the following system of differential equations:

$$\frac{d\mathbf{x}}{dt} = \begin{pmatrix} -1 & -4 & 2 \\ 0 & -1 & 3 \\ 0 & 2 & -2 \end{pmatrix} \mathbf{x}.$$

Solution: $A = \begin{pmatrix} -1 & -4 & 2 \\ 0 & -1 & 3 \\ 0 & 2 & -2 \end{pmatrix}$

The characteristic polynomial

$$\det(A - rI) = \det\begin{pmatrix} -1-r & -4 & 2 \\ 0 & -1-r & 3 \\ 0 & 2 & -2-r \end{pmatrix}$$

$$= (-1-r)[(-1-r)(-2-r) - 6]$$

$$= -(r+1)(r^2+3r-4)$$

$$= -(r+1)(r-1)(r+4) = 0$$

$$\Rightarrow r_1 = -1, \quad r_2 = 1, \quad r_3 = -4.$$

For $r_1 = -1$, we solve $(A - r_1 I)\xi = 0$

$$\begin{pmatrix} 0 & -4 & 2 \\ & 0 & 3 \\ & 2 & -1 \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} -4\xi_2 + 2\xi_3 = 0 \\ 3\xi_3 = 0 \\ 2\xi_2 - \xi_3 = 0 \end{cases} \Rightarrow \xi_2 = \xi_3 = 0 \quad \text{let } \xi_1 = c.$$

$$\Rightarrow \xi = \begin{pmatrix} c \\ 0 \\ 0 \end{pmatrix} = c\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Take $\xi^{(1)} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow X^{(1)}(t) = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{-t}$

For $r_2 = 1$, we solve $(A - r_2 I)\xi = 0$

$$\begin{pmatrix} -2 & -4 & 2 \\ 0 & -2 & 3 \\ 0 & 2 & -3 \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} -2\xi_1 - 4\xi_2 + 2\xi_3 = 0 \\ -2\xi_2 + 3\xi_3 = 0 \end{cases}$$

Let $\xi_3 = c \Rightarrow \xi_2 = \frac{3}{2}c \Rightarrow \xi_1 = -2c$

$$\Rightarrow \xi = \begin{pmatrix} -2c \\ \frac{3}{2}c \\ c \end{pmatrix} = c\begin{pmatrix} -2 \\ 3/2 \\ 1 \end{pmatrix}$$

Take $\xi^{(2)} = \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix} \Rightarrow X^{(2)}(t) = \begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix} e^{t}$

For $r_3 = -4$, we solve $(A - r_3 I)\xi = 0$

$$\begin{pmatrix} 3 & -4 & 2 \\ 0 & 3 & 3 \\ 0 & 2 & 2 \end{pmatrix}\begin{pmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} 3\xi_1 - 4\xi_2 + 2\xi_3 = 0 \\ 3\xi_2 + 3\xi_3 = 0 \end{cases}$$

Let $\xi_3 = c \Rightarrow \xi_2 = -c \Rightarrow \xi_1 = -2c$

$$\Rightarrow \xi = \begin{pmatrix} -2c \\ -c \\ c \end{pmatrix} = c\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

Take $\xi^{(3)} = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \Rightarrow X^{(3)}(t) = \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} e^{4t}$

Thus the general solution is

$$X(t) = C_1 X^{(1)}(t) + C_2 X^{(2)}(t) + C_3 X^{(3)}(t)$$

$$= C_1\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^{-t} + C_2\begin{pmatrix} -4 \\ 3 \\ 2 \end{pmatrix} e^{t} + C_3\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} e^{4t}$$